Raising power 31 on both sides,

$$(-\sqrt{3}+3i)^{31} = \left(2\sqrt{3}\right)^{31}\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)^{31}$$

$$= \left(2\sqrt{3}\right)^{31}\left(\cos\left(20\pi+\frac{2\pi}{3}\right)+i\sin\left(20\pi+\frac{2\pi}{3}\right)\right)$$

$$= \left(2\sqrt{3}\right)^{31}\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)$$

$$= \left(2\sqrt{3}\right)^{31}\left(\cos\left(\pi-\frac{\pi}{3}\right)+i\sin\left(\pi-\frac{\pi}{3}\right)\right)$$

$$= \left(2\sqrt{3}\right)^{31}\left(-\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right) = \left(2\sqrt{3}\right)^{31}\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right).$$

2.8.2 Finding n^{th} roots of a complex number

de Moivre's formula can be used to obtain roots of complex numbers. Suppose n is a positive integer and a complex number ω is n^{th} root of z denoted by $z^{1/n}$, then we have

$$\omega^n = z. \qquad \ldots(1)$$

Let $\omega = \rho(\cos\phi + i\sin\phi)$ and

$$z = r(\cos\theta + i\sin\theta) = r\left(\cos(\theta+2k\pi)+i\sin(\theta+2k\pi)\right),\ k\in\mathbb{Z}$$

Since w is the nth root of z, then

$$\omega^n = z$$

$$\Rightarrow\ \rho^n(\cos\phi+i\sin\phi)^n = r\left(\cos(\theta+2k\pi)+i\sin(\theta+2k\pi)\right),\ k\in\mathbb{Z}$$



By de Moivre's theorem,

$$\rho^n(\cos n\phi + i\sin n\phi) = r\left(\cos(\theta+2k\pi)+i\sin(\theta+2k\pi)\right),\ k\in\mathbb{Z}$$

Comparing the moduli and arguments, we get

$$\rho^n = r \text{ and } n\phi = \theta+2k\pi,\ k\in\mathbb{Z}$$

$$\rho = r^{1/n} \text{ and } \phi = \frac{\theta+2k\pi}{n}, k\in\mathbb{Z}.$$

Therefore, the values of ω are $r^{1/n}\left(\cos\left(\frac{\theta+2k\pi}{n}\right)+i\sin\left(\frac{\theta+2k\pi}{n}\right)\right),\ k\in\mathbb{Z}$.

Although there are infinitely many values of k, the distinct values of ω are obtained when $k = 0,1,2,3,\ldots,n-1$. When $k = n, n+1, n+2,\ldots$ we get the same roots at regular intervals (cyclically). Therefore the nth roots of complex number $z = r(\cos\theta + i\sin\theta)$ are

$$z^{1/n} = r^{1/n}\left(\cos\left(\frac{\theta+2k\pi}{n}\right)+i\sin\left(\frac{\theta+2k\pi}{n}\right)\right),\ k = 0,1,2,3,\ldots,n-1.$$

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